NESA NATIONAL EXAMINATION AND SCHOOL INSPECTION AUTHORITY

SOD– Web Application Development

T171

Monday, 29/07/2024

08:30 – 11:30 AM

Names:.....

Index number :.....

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2023-2024

QUESTIONS and ANSWERS BOOKLET

OPTION/ TRADE : SOFTWARE DEVELOPMENT

SUBJECT/EXAM: WEB APPLICATION DEVELOPMENT

DURATION: 3 Hours

Read carefully the instructions on page 1 & 2.

TOTAL MARKS...../100

FOR EXAMINER'S USE ONLY

QUESTIONS	1	2	3	4	5	6	7	8	9	10	Total
Marks											

QUESTIONS	11	12	13	14	15	16	17	18	19	20	Total
Marks											

QUESTIONS	21	22	23	24	25	26	27	28	29	30	Total
Marks											

TSS NATIONAL EXAMINATIONS, LEVEL 5, 2023-2024

INSTRUCTIONS TO CANDIDATES (ANSWER BOOKLET)

1. A candidate should fill in the actual names and the Index number on the cover of this questions and answer booklet on the provided place.

2. It is illegal for a candidate to write any of names, Index number or school name inside the answer booklet.

3. No candidate should remove or tear any pages or part of it in the answer booklet.

4. A candidate should answer in the language in which the examination is set.

5. A candidate should sign on the sitting plan when submitting the answer booklet. He/she has also to check if the answer booklet is well sealed.

6. No extra paper is allowed in the examinations room. If a candidate is caught with it his/her results will be nullified.

7. No candidate is allowed to write answers not related to the subject being sat for, otherwise it will be considered as a cheating case.

8. Write your answers on the 16 lined pages (From page 7 to page 22).

9. Use the last non-lined pages as draft.

10. Results for any candidate who is caught in examination malpractices are nullified. The cheating can be recognized during examinations administration, marking exercise or even thereafter.

N.B: 1) After results publication, there is no remarking and no candidate is given his/her answer booklet for review. This answer booklet is a property of NESA.

2) Claims are only received online within 30 days after results publication. A link will be provided after results publication.

T171 Web Application Development

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INSTRUCTIONS TO CANDIDATES (QUESTION PAPER)

This Exam paper is composed of Three Sections (A, B, and C). Follow the instructions given below, and answer the indicated questions for a total of 100 marks

Section **A**: Fourteen (**14**) questions, all **Compulsory**	**55 marks**
Section **B**: Among the five (**5**) questions, attempt any three (3)	**30 marks**
Section **C**: Among the two (**2**) questions, attempt any one (1)	**15 marks**

Allowed materials:

- Ruler
- Blue or black pen
- Calculator

Note:

Every candidate is required to carefully comply with the provided assessment instructions.

T171 Web Application Development

SECTION A: Attempt all questions **(55 marks)**

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01. Match the following PHP functions with their descriptions: **(5 marks)**

Functions	Descriptions
1. Mysqli_connect()	a) Returns the number of rows affected by the last query
2. Mysqli_query()	b) Retrieves a row of data as an associative or numeric array
3. Mysqli_num_rows()	c) Closes the connection to the MySQL server
4. Mysqli_fetch_array()	d) Executes a SQL query on the selected database.
5. Mysqli_close ()	e) Establishes a connection to the MySQL server

02. Give at least five (5) advantages of using a PHP framework for web development. **(5 marks)**

03. List three (3) web application cloud deployment models. **(3 marks)**

04. Which SQL statement among the following is used to delete records from a table based on a specified condition? **(3 marks)**

a. INSERT INTO

b. UPDATE

c. SELECT

d. DELETE FROM

05. What is the purpose of using a PHP framework in web development? **(3 marks)**

06. List any four (4) types of iterative statements used in PHP. **(4 marks)**

07. Discuss the key features and components provided by Phalcon for building web applications. **(5 marks)**

08. Differentiate scripting language from programming language. **(4 marks)**

09. Outline any two (2) examples of PHP Framework. **(2 marks)**

10. Explain the role of each of the following components in the authentication process: **(4 marks)**

i. Server,

ii. User ID,

iii. Password,

iv. Database name

11. What will be the output of the following program? **(5 marks)**

```
 <html>
<body>
    <?php
      $a = 0;
      $b = 0;
      for( $i = 0; $i<5; $i++ ) {
          $a += 50;
          $b += 15;
     }
     echo ("At the end of the loop a = $a and
b = $b");
       ?>
 </body>
</html>
```

12. Interpret the following PHP codes and write their output. **(3 marks)**

```
<?php
$num1 = 10;
$num2 = 5;
$result = $num1 + $num2 * 2;
echo "The result is: $result";
?>
```

13. Assume you are connecting to Apache local server with x_db database. Use mysqli_connect () object to connect to this server with no password. **(5 marks)**

14. Identify and correct the error in the following PHP code, rewrite the update version of the code with no error. **(4 marks)**

```
$x = 5;
$y = 10;
if ($x = $y) {
    echo "x is equal to y";
} elseif ($x > $y) {
    echo "x is greater than y";
} else {
    echo "x is less than y";
}
```

Section B: Attempt any three (3) questions **(30 marks)**

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15. Explain any five (5) Security Implications of Using a PHP Framework for Web Application Development. **(10 marks)**

16. To integrate the mydata database with an HTML form using PHP, several parameters need to be set to establish a connection and interact with the database effectively. with the help of example assess the parameters required. **(10 marks)**

17. Evaluate the significance of syntax errors, logical errors, and runtime errors in PHP programming based on their distinct characteristics and impacts on program execution. **(10 marks)**

18. Compare functional requirement and non-functional requirement in web application deployment. **(10 marks)**

19. How does the switch statement differ from if statement in PHP? **(10 marks)**

Section C: Attempt only one (1) question **(15 marks)**

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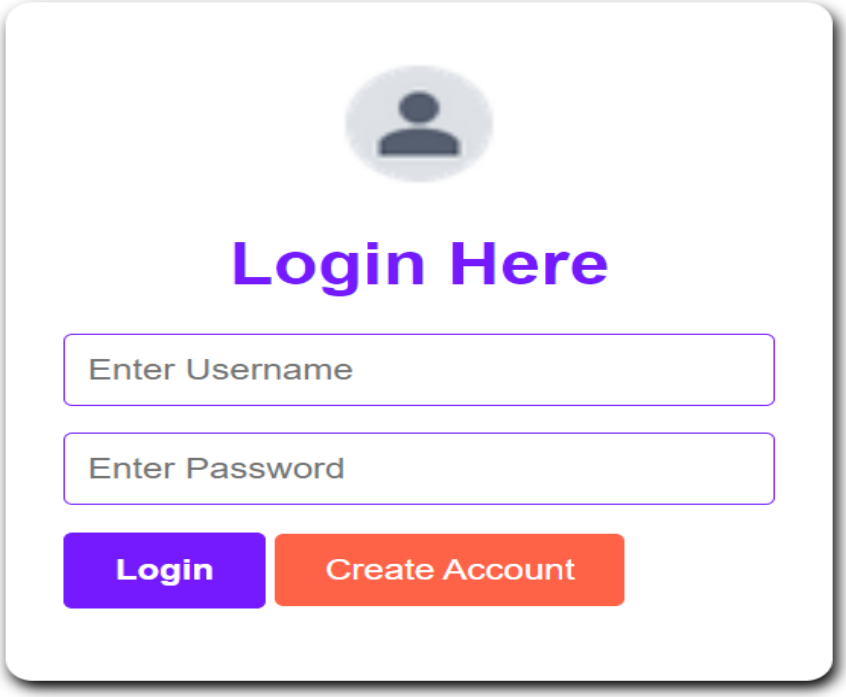
20. Write PHP code to perform the following login form. Suppose that database name is "**motive**" and table name is "**users**". **(15 marks)**



21. Use the following information given below: **(15 marks)**

Database: schools

Table: students with the following columns (stid, fullname varchar (255), sex char (1), age int (11), Address varchar (255)).

Create PHP scripts to display the female students who have the age between 18 and 21.

END OF ASSESSMENT

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